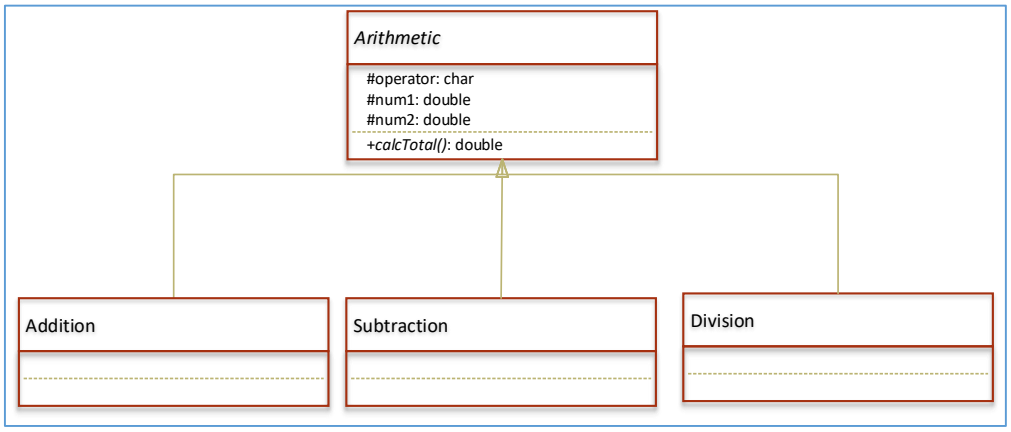
Exercise: Factory pattern

Question 1

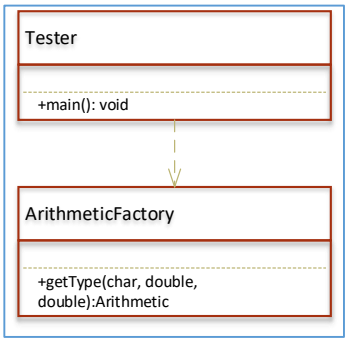
The UML diagram shows the classes **Arithmetic**, **Addition**, **Subtract** and **Division**. The subclasses override **calcTotal**() as follows, if **operator** is:

- '+': ret urn num1 + num2 in **Addition**
- '-': return num1 - num2 in **Subtraction**
- '/': returns num1 + num2 in **Addition**

1. Write the code for **Arithmetic, Addition** and Subtraction and Division



2. Write the factory class **ArithmeticFactory** below. The method **getType**():
   a. Takes three parameters (an arithmetic operator and two numbers) and
   b. Based on the operator, returns an instance of a subclass of **Arithmetic** (e.g., an instance of Addition if the symbol is '+'; Subtraction if the symbol is '-' , etc.).



3. Write the class **Tester** above. The code:
   a. Creates an instance of the factory class **ArithmeticFactory**
   b. Calls **getType**() with parameters to return an instance of the appropriate subclass of **Arithmetic**
   c. Displays the instance returned in (b). A typical output is shown below.

```
operator: +
num1= 30.0 and num2=20.0 total = 50.0
operator: -
num1= 40.0 and num2=50.0 total = -10
```

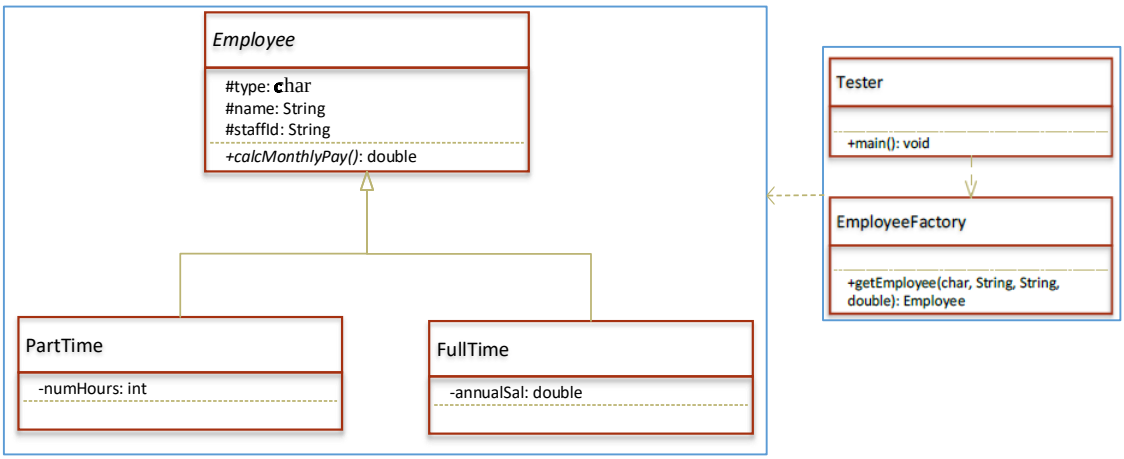
Question 2

The UML diagram shows the classes **Employee**, **PartTime** and **FullTime**, **EmployeeFactory** and **Tester**.

The method **calcMonthlyPay**() returns:

- numHours * 80 in PartTime
- 1/12*annualSal in FullTime

1. Write the code for the **Employee**, **PartTime** and **FullTime**.



2. Write the factory class **EmployeeFactory** below. The method **getEmployee**():
   a. Takes four parameters: a character (**type**: i.e., type of employee), a string **(staffId:** an employee's ID); a string (**name**: an employee's name) and a number (the number of hours worked if PartTime or annual salary, if FullTime).
   b. Returns an instance of a **PartTime** (if the character is 'P' or 'p') or FullTime (if the character is 'F' or 'f').

4. Write the class **Tester** above. The code:
   a. Creates an instance of the factory class **EmployeeFactory**
   b. Calls **getEmployee**() to return an instance of the appropriate subclass and
   c. Displays the instance returned. A typical output is shown below:

```
Type: F
Employee Id: JACAN002
Name: Jane Cavani
Annual salary: 120000.0
Pay: 10000.0

Type: P
Employee Id: KOMEN001
Name: Kofi Mensah
No. of hours: 120
Rate: 80
Pay: 9600.0
```